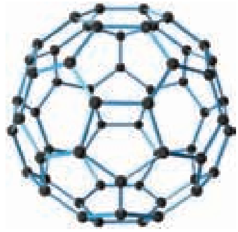
and Sir Harry Kroto at the University of Sussex discovered C60, a carbon nanoparticle shaped like a soccer ball. The unique molecule was named Buckminsterfullerene after the visionary American architect and engineer Buckminster Fuller who designed the geodesic dome. More commonly called a "buckyball", the molecule is extremely rugged, capable of surviving collisions with metals and other materials at speeds higher than 20,000 miles per hour.


"Elusive" Buckyballs

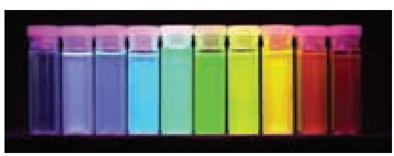
Atomic Force Microscope (1986)

Invented by Gerg Binnig and his colleague Christoph Gerber at IBM, San Jose, and Calvin Quate at Stanford University, the atomic force microscope or AFM, has the capability to view, measure, and manipulate materials down to fractions of a nano in size, including measurement of various forces intrinsic to nanomaterials. The AFM can make 3-D images of an object's surface topography with extremely high magnifications (up to 1,000,000 times).


Atomic Force Microscope

Discovery of Quantum Dots (1988)

In the early 1980s, Dr. Louis Brus and his team of researchers at Bell Laboratories made a significant contribution to the field of nanotechnology when they discovered that nano-sized crystal semiconductor materials made from the same substance exhibited strikingly different colours. These nanocrystal semiconductors were called quantum dots and this work eventually contributed to the understanding of the Quantum Confinement Effect, which explains the relationship between size and colour for these nanocrystals. Scientists have learned how to control the size of quantum dots making it possible to obtain a broad range of colours. Quantum dots have the potential to revolutionise the way solar energy is collected, improve medical diagnostics by


Different colours of quantum dots